



Read Bit's Stack

Quick facilitation notes & answer key

Goal

Children trace (run) a short block program on a 1D stage and report the result — Bit's landing box. Running each block in order, one at a time, is executing a program; the number and direction of the blocks decide where Bit ends up.

Kindergarten frame (Ontario)

Problem Solving and Innovating — children carry out a sequence of coded instructions and determine the outcome.

No reading required of the child: read each prompt aloud; the child writes the landing-box number (1-5).

Materials

The printed worksheet and a pencil. Optional: floor boxes 1-5 with a star on box 4 so a child can walk each stack.

Run it (about 15 min)

1. MODEL: put a finger on box 1. Run Stack 1 aloud — '➡ box 2... ➡ box 3.' Bit lands on box 3. One finger-move per block, top to bottom.
2. GUIDED: Exercise 1 — children run Stack 1 to box 3. Exercise 2 — Stack 2 is one more ➡, landing on box 4, the star.
3. YOU TRY: Exercise 3 — Stack 3 ends with ⬅, so after ➡ ➡ to box 3 the last block steps back to box 2.
4. Connect: 'To find out what a program does, you run it — one block at a time.'

Answer key (modelled by the run-gate)

Ex 1 — Stack 1 (➡ ➡) from box 1 → box 3.

Ex 2 — Stack 2 (➡ ➡ ➡) from box 1 → box 4, yes, the star.

Ex 3 — Stack 3 (➡ ➡ ⬅) from box 1 → box 3, then back to box 2. Bit lands on box 2.

Watch for & differentiate

Common slip: counting box 1 as the first move — the first ➡ moves Bit OFF box 1 onto box 2.

Watch Ex 3: children may forget the last ⬅ steps backward — have them touch each box as they go.

Simplify: do Stacks 1-2 only (all ➡). Skip the ⬅ in Stack 3 for children who need it.

Extend: ask which stack lands Bit on the star, and how they could change Stack 1 to reach it (add one ➡).

Movement option: children physically walk each stack on the floor boxes.

Success looks like

The child runs each stack one block at a time and writes box 3, box 4 (star), and box 2 — tracking the ⬅ step back in Stack 3.